

GIUSEPPE SCOTTI

associate professor

基本信息

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学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

低功耗电路与能量管理

模拟与射频IC设计

混合信号集成技术

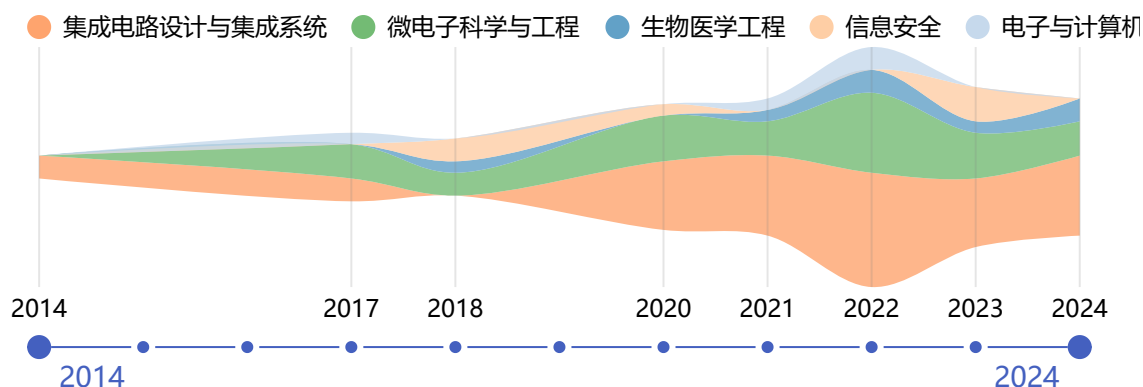
模拟与数模混合电路

数字集成电路设计

个人简介

Giuseppe Scotti, born in Cagliari, Italy, in 1975, holds M.S. and Ph.D. degrees in electronic engineering from the University of Rome "La Sapienza." He is currently an Associate Professor at the DIET department of the University of Rome "La Sapienza," teaching courses on electronics and microelectronics. His research focuses on integrated circuits design, particularly robust design methodologies for analog and digital VLSI circuits. Scotti has coauthored over 45 journal publications, 70 conference papers, and is a co-inventor of 2 international patents. He has extensive experience in both analog and digital integrated circuits design using Cadence and Synopsys tools.

近年发表的论文



An Ultra-Low-Voltage Approach to Accurately Set the Quiescent Current of Digital Standard Cells Used for Analog Design and Its Application on an Inverter-Based Operational Transconductance Amplifier

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Della Sala, Riccardo; Centurelli, Francesco; Scotti, Giuseppe

被引用次数: 0

2024

A 0.3 V OTA with Enhanced CMRR and High Robustness to PVT Variations

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Della Sala, Riccardo; Centurelli, Francesco; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 0

2024

A Novel Technique to Design Ultra-Low Voltage and Ultra-Low Power Inverter-Based OTAs

2024 19TH CONFERENCE ON PH.D RESEARCH IN MICROELECTRONICS AND ELECTRONICS, PRIME 2024

Della Sala, Riccardo; Centurelli, Francesco; Scotti, Giuseppe

被引用次数: 0

2024

A 150 MS/s, 10 bit SAR ADC Featuring a Modified Quasi-Monotonic Switching Scheme

2024 19TH CONFERENCE ON PH.D RESEARCH IN MICROELECTRONICS AND ELECTRONICS, PRIME 2024

Spinogatti, Valerio; Bocciarelli, Cristian; Eusebio, Lorenzo; Centurelli, Francesco; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 0

2024

On the Feasibility of Cascode and Regulated Cascode Amplifier Stages in ULV Circuits Exploiting MOS Transistors in Deep Subthreshold Operation

IEEE ACCESS

Della Sala, Riccardo; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe

被引用次数: 1

2024

A Novel High Performance Standard-Cell Based ULV OTA Exploiting an Improved Basic Amplifier

IEEE ACCESS

Della Sala, Riccardo; Centurelli, Francesco; Scotti, Giuseppe

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2024

Rail to Rail ICMR and High Performance ULV Standard-Cell-Based Comparator for Biomedical and IoT Applications

IEEE ACCESS

Della Sala, Riccardo; Centurelli, Francesco; Scotti, Giuseppe; Palumbo, Gaetano

被引用次数: 8

2024

A 0.064 mm² 16-Channel In-Pixel Neural Front End with Improved System Common-Mode Rejection Exploiting a Current-Mode Summing Approach

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Nicolini, Giovanni; Fava, Alessandro; Centurelli, Francesco; Scotti, Giuseppe

被引用次数: 0

2024

Exploiting Body-Driven Feedbacks in Physical Unclonable Functions for Ultra Low Voltage, Ultra Low Power Applications: A 0.3 V Weak-PUF

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Della Sala, Riccardo; Bellizia, Davide; Centurelli, Francesco; Scotti, Giuseppe; Trifiletti, Alessandro

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2024

A Monostable Physically Unclonable Function Based on Improved RCCMs with 0-1.56% Native Bit Instability at 0.6-1.2 V and 0-75 °C

ELECTRONICS

Della Sala, Riccardo; Bellizia, Davide; Centurelli, Francesco; Scotti, Giuseppe

被引用次数: 6

2023

A Novel FPGA Implementation of the NAND-PUF with Minimal Resource Usage and High Reliability

CRYPTOGRAPHY

Della Sala, Riccardo; Scotti, Giuseppe

被引用次数: 8

2023

A Detailed Model of Cyclostationary Noise in Switched-Resistor Circuits

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Fava, Alessandro; Centurelli, Francesco; Scotti, Giuseppe

被引用次数: 2

2023

High-Accuracy Low-Cost Generalized Complex Pruned Volterra Models for Nonlinear Calibration

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Bocciarelli, Cristian; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Spinogatti, Valerio; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 1

2023

A High Performance 0.3 V Standard-Cell-Based OTA Suitable for Automatic Layout Flow

APPLIED SCIENCES-BASEL

Della Sala, Riccardo; Centurelli, Francesco; Scotti, Giuseppe

A body-driven rail-to-rail 0.3 V operational transconductance amplifier exploiting current gain stages

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Della Sala, Riccardo; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

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2023

A 0.3V Rail-to-Rail Three-Stage OTA With High DC Gain and Improved Robustness to PVT Variations

IEEE ACCESS

Della Sala, Riccardo; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 14

2023

Exploiting the DD-Cell as an Ultra-Compact Entropy Source for an FPGA-Based Re-Configurable PUF-TRNG Architecture

IEEE ACCESS

Sala, Riccardo Della; Scotti, Giuseppe

被引用次数: 7

2023

Simple and Accurate Model for the Propagation Delay in MCML Gates

ELECTRONICS

Giustolisi, Gianluca; Scotti, Giuseppe; Palumbo, Gaetano

被引用次数: 0

2023

A Novel Ultra-Compact FPGA-Compatible TRNG Architecture Exploiting Latched Ring Oscillators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Della Sala, Riccardo; Bellizia, Davide; Scotti, Giuseppe

被引用次数: 38

2022

A Lightweight FPGA Compatible Weak-PUF Primitive Based on XOR Gates

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Della Sala, Riccardo; Bellizia, Davide; Scotti, Giuseppe

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2022

Enabling ULV Fully Synthesizable Analog Circuits: The BA Cell, a Standard-Cell-Based Building Block for Analog Design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Della Sala, Riccardo; Centurelli, Francesco; Scotti, Giuseppe

被引用次数: 17

2022

Methods for Model Complexity Reduction for the Nonlinear Calibration of Amplifiers Using Volterra Kernels

ELECTRONICS

Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 1

2022

A Differential-to-Single-Ended Converter Based on Enhanced Body-Driven Current Mirrors Targeting Ultra-Low-Voltage OTAs

ELECTRONICS

Sala, Riccardo Della; Centurelli, Francesco; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 11

2022

A Tree-Based Architecture for High-Performance Ultra-Low-Voltage Amplifiers

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Centurelli, Francesco; Della Sala, Riccardo; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 19

2022

A Standard-Cell-Based CMFB for Fully Synthesizable OTAs

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Centurelli, Francesco; Della Sala, Riccardo; Scotti, Giuseppe

被引用次数: 18

2022

A SiGe HBT 6th-Order 10 GHz Inductor-Less Anti-Aliasing Low-Pass Filter for High-Speed ATI Digitizers

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 7

2022

High-Throughput FPGA-Compatible TRNG Architecture Exploiting Multistimuli Metastable Cells

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Della Sala, Riccardo; Bellizia, Davide; Scotti, Giuseppe

被引用次数: 21

2022

A Biasing Approach to Design Ultra-Low-Power Standard-Cell-Based Analog Building Blocks for Nanometer SoCs

IEEE ACCESS

Centurelli, Francesco; Giustolisi, Gianluca; Pennisi, Salvatore; Scotti, Giuseppe

被引用次数: 11

2022

A Novel Differential to Single-Ended Converter for Ultra-Low-Voltage Inverter-Based OTAs

IEEE ACCESS

Della Sala, Riccardo; Centurelli, Francesco; Scotti, Giuseppe

被引用次数: 17

2022

A Novel Clock Gating Approach for the Design of Low-Power Linear Feedback Shift Registers

IEEE ACCESS

Giustolisi, Gianluca; Mita, Rosario; Palumbo, Gaetano; Scotti, Giuseppe

被引用次数: 3

2022

Sub- μ W Front-End Low Noise Amplifier for Neural Recording Applications

PRIME 2022: 17TH INTERNATIONAL CONFERENCE ON PHD RESEARCH IN MICROELECTRONICS AND ELECTRONICS

Della Sala, Riccardo; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe

被引用次数: 2

2022

High-efficiency 0.3V OTA in CMOS 130nm technology using current mirrors with gain

PRIME 2022: 17TH INTERNATIONAL CONFERENCE ON PHD RESEARCH IN MICROELECTRONICS AND ELECTRONICS

Della Sala, Riccardo; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe

被引用次数: 2

2022

80 dB tuning range transimpedance amplifier exploiting the Switched-Resistor approach

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Centurelli, Francesco; Fava, Alessandro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 4

2022

Klessydra-T: Designing Vector Coprocessors for Multithreaded Edge-Computing Cores

IEEE MICRO

Cheikh, Abdallah; Sordillo, Stefano; Mastrandrea, Antonio; Menichelli, Francesco; Scotti, Giuseppe; Olivieri, Mauro

被引用次数: 17

2021

A 0.3 V, Rail-to-Rail, Ultralow-Power, Non-Tailed, Body-Driven, Sub-Threshold Amplifier

APPLIED SCIENCES-BASEL

Centurelli, Francesco; Della Sala, Riccardo; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 24

2021

An E-band Variable Gain Amplifier with 24 dB-control range and 80 to 100 GHz 1 dB bandwidth in SiGe BiCMOS technology

FREQUENZ

Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 1

2021

0.5-V Frequency Dividers in Folded MCML Exploiting Forward Body Bias: Analysis and Comparison

ELECTRONICS

Centurelli, Francesco; Scotti, Giuseppe; Palumbo, Gaetano

被引用次数: 2

2021

A Novel OTA Architecture Exploiting Current Gain Stages to Boost Bandwidth and Slew-Rate

ELECTRONICS

Centurelli, Francesco; Della Sala, Riccardo; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 13

2021

A Novel Standard-Cell-Based Implementation of the Digital OTA Suitable for Automatic Place and Route

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Palumbo, Gaetano; Scotti, Giuseppe

被引用次数: 16

2021

A Very-Low-Voltage Frequency Divider in Folded MOS Current Mode Logic With Complementary n- and p-type Flip-Flops

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Centurelli, Francesco; Scotti, Giuseppe; Palumbo, Gaetano

被引用次数: 8

2021

Distributed switched-resistor approach for high-Q biquad filters

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Centurelli, Francesco; Fava, Alessandro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 5

2021

Design of Low-Voltage Power Efficient Frequency Dividers in Folded MOS Current Mode Logic

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Centurelli, Francesco; Scotti, Giuseppe; Trifiletti, Alessandro; Palumbo, Gaetano

被引用次数: 5

2021

An improved reversed miller compensation technique for three-stage CMOS OTAs with double pole-zero cancellation and almost single-pole frequency response

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 3

2020

Delay models and design guidelines for MCML gates with resistor or PMOS load

MICROELECTRONICS JOURNAL

Centurelli, Francesco; Scotti, Giuseppe; Trifiletti, Alessandro; Palumbo, Gaetano

被引用次数: 7

2020

0.6-V CMOS cascode OTA with complementary gate-driven gain-boosting and forward body bias

INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS

Cellucci, Danilo; Centurelli, Francesco; Di Stefano, Valerio; Monsurro, Pietro; Pennisi, Salvatore; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 19

2020

A Power Efficient Frequency Divider With 55 GHz Self-Oscillating Frequency in SiGe BiCMOS

ELECTRONICS

Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Tommasino, Pasquale; Trifiletti, Alessandro

被引用次数: 3

2020

A Novel 0.5 V MCML D-Flip-Flop Topology Exploiting Forward Body Bias Threshold Lowering

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Scotti, Giuseppe; Trifiletti, Alessandro; Palumbo, Gaetano

被引用次数: 17

2020

A Multi-Folded MCML for Ultra-Low-Voltage High-Performance in Deeply Scaled CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Palumbo, Gaetano; Scotti, Giuseppe

被引用次数: 5

2020

SC-DDPL: A Novel Standard-Cell Based Approach for Counteracting Power Analysis Attacks in the Presence of Unbalanced Routing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Bellizia, Davide; Bongiovanni, Simone; Olivieri, Mauro; Scotti, Giuseppe

被引用次数: 19

2020

Secure Double Rate Registers as an RTL Countermeasure Against Power Analysis Attacks

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Bellizia, Davide; Bongiovanni, Simone; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro; Trotta, Francesco Bruno

被引用次数: 30

2018

Low power DDA-based instrumentation amplifier for neural recording applications in 65 nm CMOS

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Avoli, Matteo; Centurelli, Francesco; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 26

2018

TEL Logic Style as a Countermeasure Against Side-Channel Attacks: Secure Cells Library in 65nm CMOS and Experimental Results

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Bellizia, Davide; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 13

2018

Univariate Power Analysis Attacks Exploiting Static Dissipation of Nanometer CMOS VLSI Circuits for Cryptographic Applications

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Bellizia, Davide; Bongiovanni, Simone; Monsurro, Pietro; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 15

2017

A Novel Framework to Estimate the Path Delay Variability On the Back of an Envelope via the Fan-Out-of-4 Metric

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Alioto, Massimo; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 22

2017

0.9-V Class-AB Miller OTA in 0.35- μ m CMOS With Threshold-Lowered Non-Tailed Differential Pair

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Grasso, Alfio Dario; Pennisi, Salvatore; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 48

2017

Design of Low-Voltage High-Speed CML D-Latches in Nanometer CMOS Technologies

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Scotti, Giuseppe; Bellizia, Davide; Trifiletti, Alessandro; Palumbo, Gaetano

被引用次数: 24

2017

88- μ A 1-MHz Stray-Insensitive CMOS Current-Mode Interface IC for Differential Capacitive Sensors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Scotti, Giuseppe; Pennisi, Salvatore; Monsurro, Pietro; Trifiletti, Alessandro

被引用次数: 41

2014

Effectiveness of Leakage Power Analysis Attacks on DPA-Resistant Logic Styles Under Process Variations

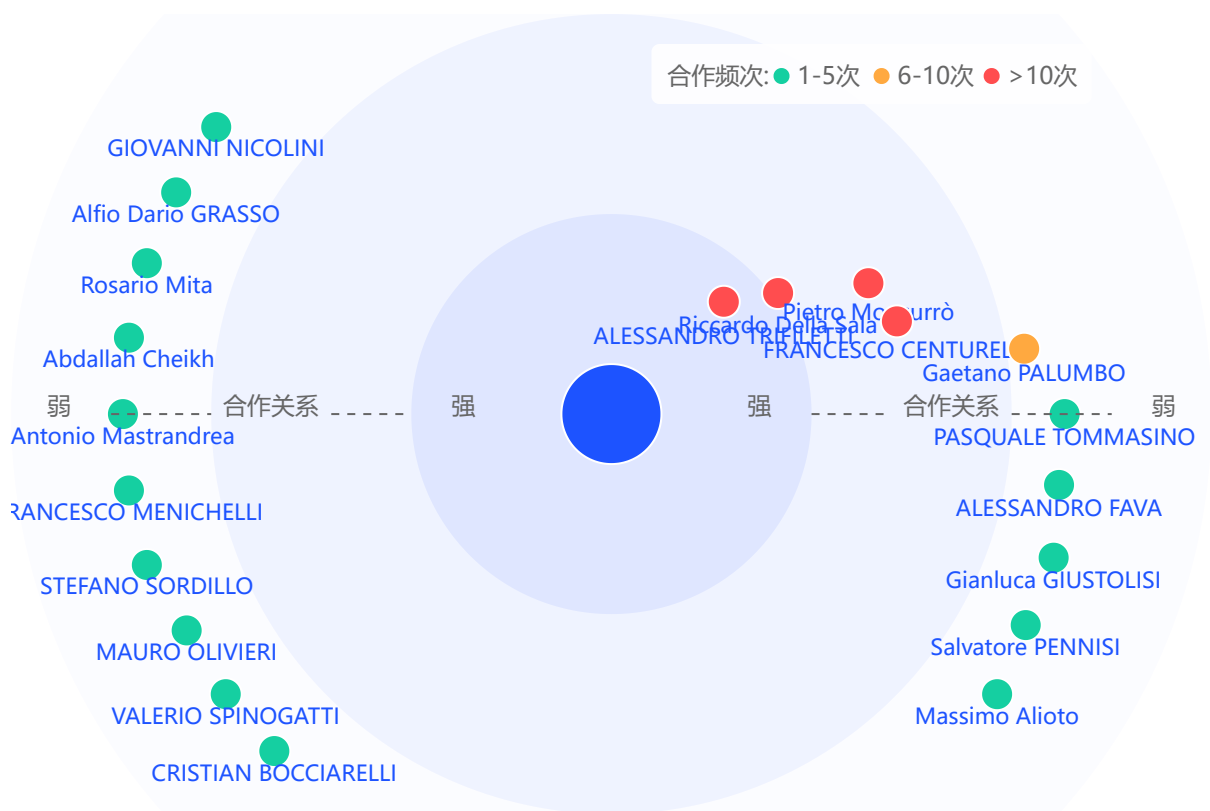
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Alioto, Massimo; Bongiovanni, Simone; Djukanovic, Milena; Scotti, Giuseppe; Trifiletti, Alessandro

被引用次数: 55

2014

学术合作



ALESSANDRO TRIFILETTI

罗马第一大学

合作次数: 29

Riccardo Della Sala

罗马第一大学

合作次数: 25

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